



Introducing ORCID

 JOHN BOHANNON AND KIRK DORAN [Authors Info & Affiliations](#)

SCIENCE • 19 May 2017 • Vol 356, Issue 6339 • pp. 691–692 • DOI:10.1126/science.356.6339.691

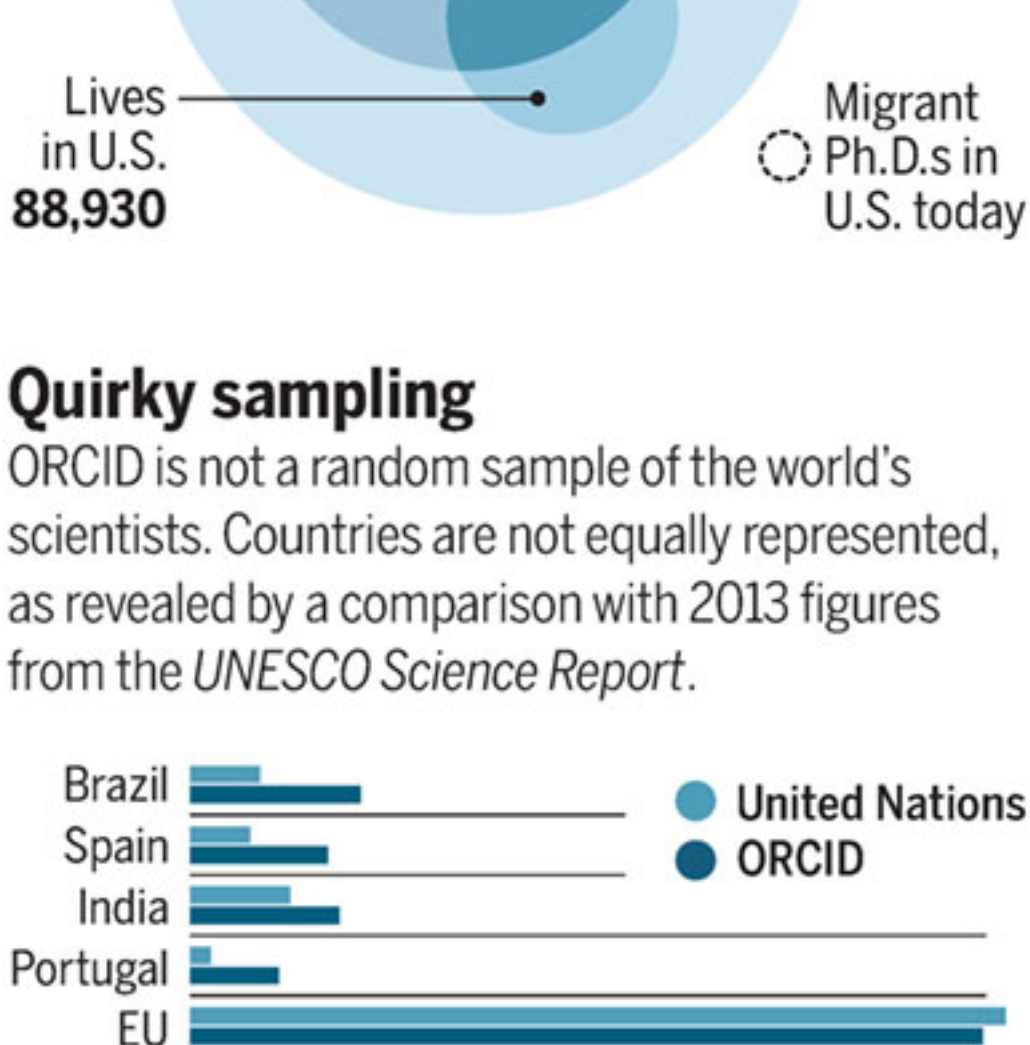
PDF

460

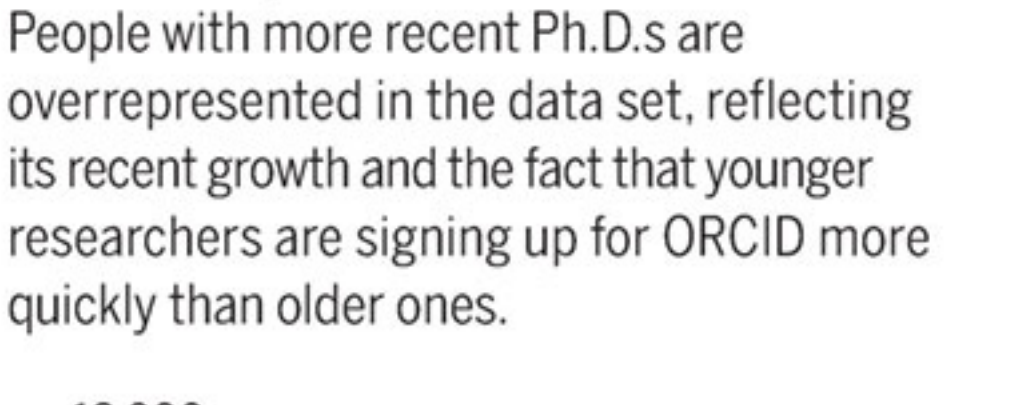


Revealing subsets

ORCID users voluntarily have created a giant and growing database.

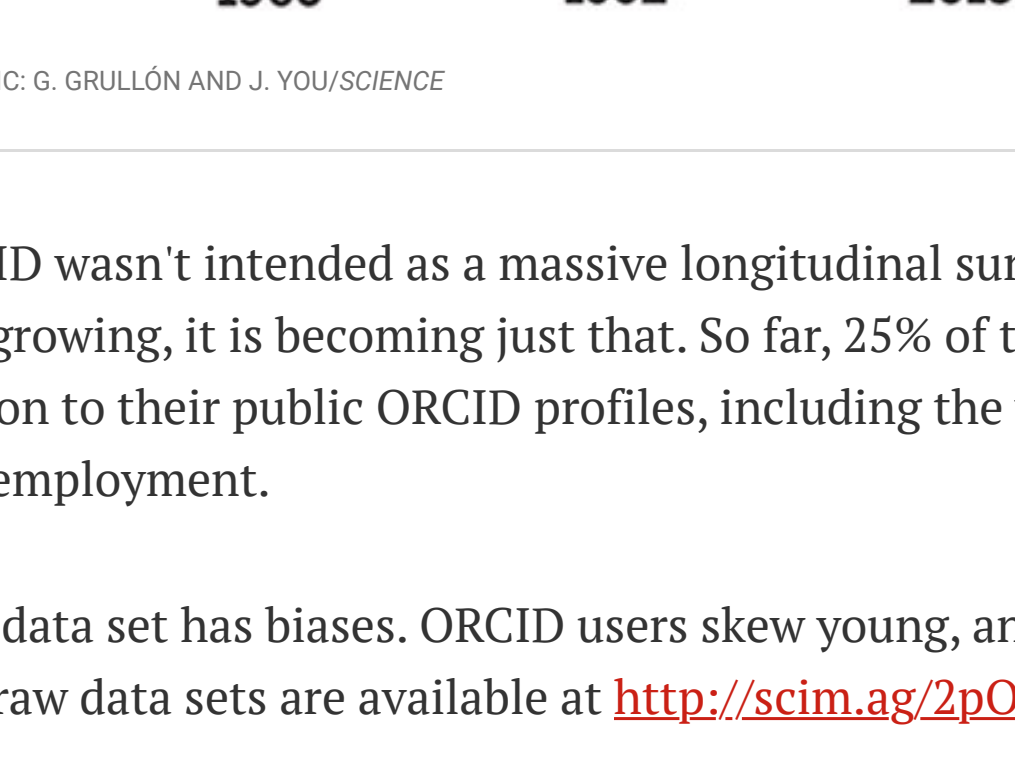


Quirky sampling

 ORCID is not a random sample of the world's scientists. Countries are not equally represented, as revealed by a comparison with 2013 figures from the *UNESCO Science Report*.


Youthful picture

People with more recent Ph.D.s are overrepresented in the data set, reflecting its recent growth and the fact that younger researchers are signing up for ORCID more quickly than older ones.



GRAPHIC: G. GRULLÓN AND J. YOU/SCIENCE

ORCID wasn't intended as a massive longitudinal survey of human migration, but with 3 million profiles and growing, it is becoming just that. So far, 25% of the world's researchers have added personal information to their public ORCID profiles, including the years, locations, and descriptions of their education and employment.

 This data set has biases. ORCID users skew young, and certain countries are over- and underrepresented. The raw data sets are available at <http://scim.ag/2pO820I>.

Scientific migrants: a view from ORCID

SIGN UP FOR THE SCIENCE eTOC

 Get the latest table of contents from *Science* delivered right to you!

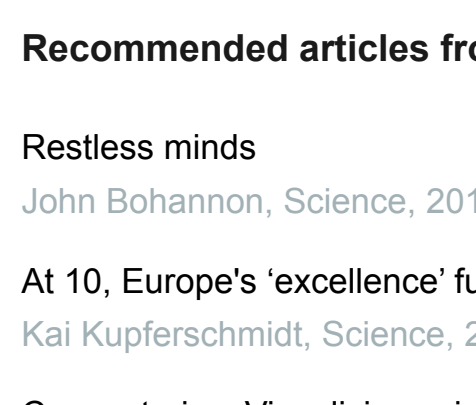
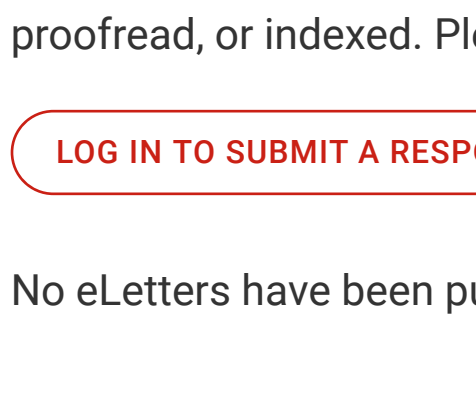
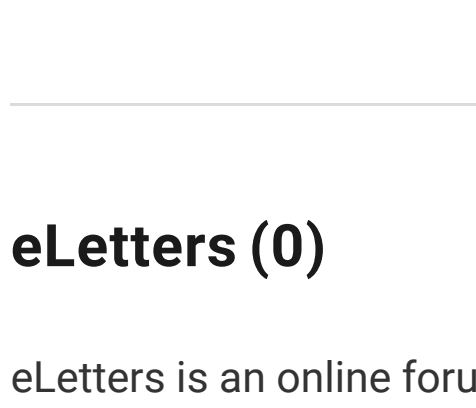
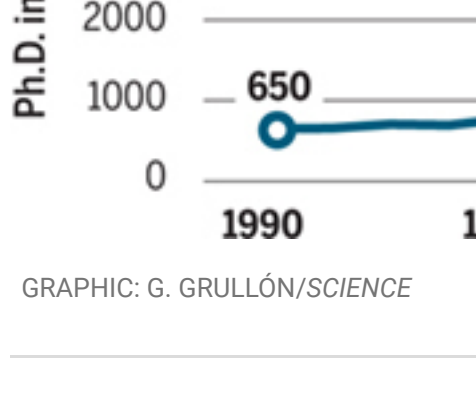
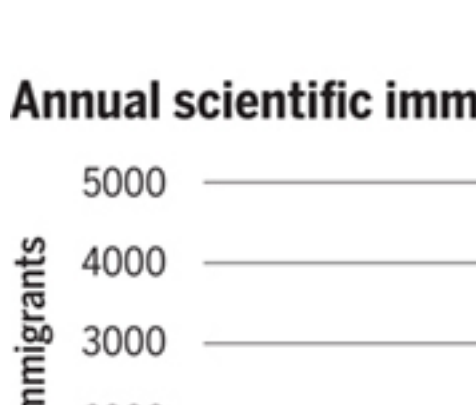
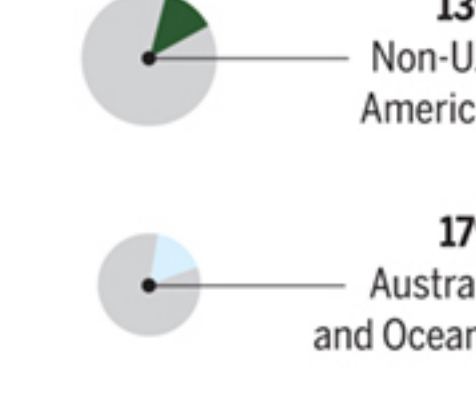
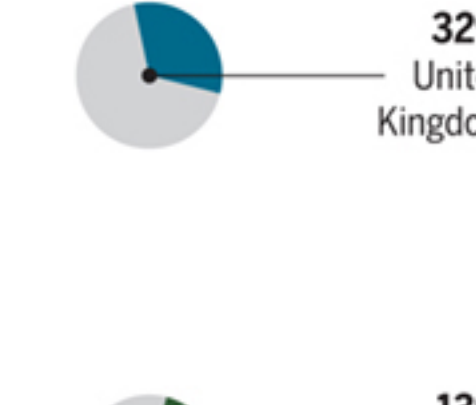
[SIGN UP >](#)

 Although the data set overrepresents some EU countries and underrepresents China, *Science's* analysis reveals intriguing migration patterns. For example, about a third of those who earned their Ph.D. in the United Kingdom were living in another country by 2016. But only about 15% of Ph.D.s from other EU nations migrated away. Also, ORCID chronicles steady growth since 1990 in the number of foreign scientists immigrating to the United States (bottom). But in 2002, that annual influx stagnated, possibly because of the 2001 terrorist attacks.

Number of people who earned Ph.D.



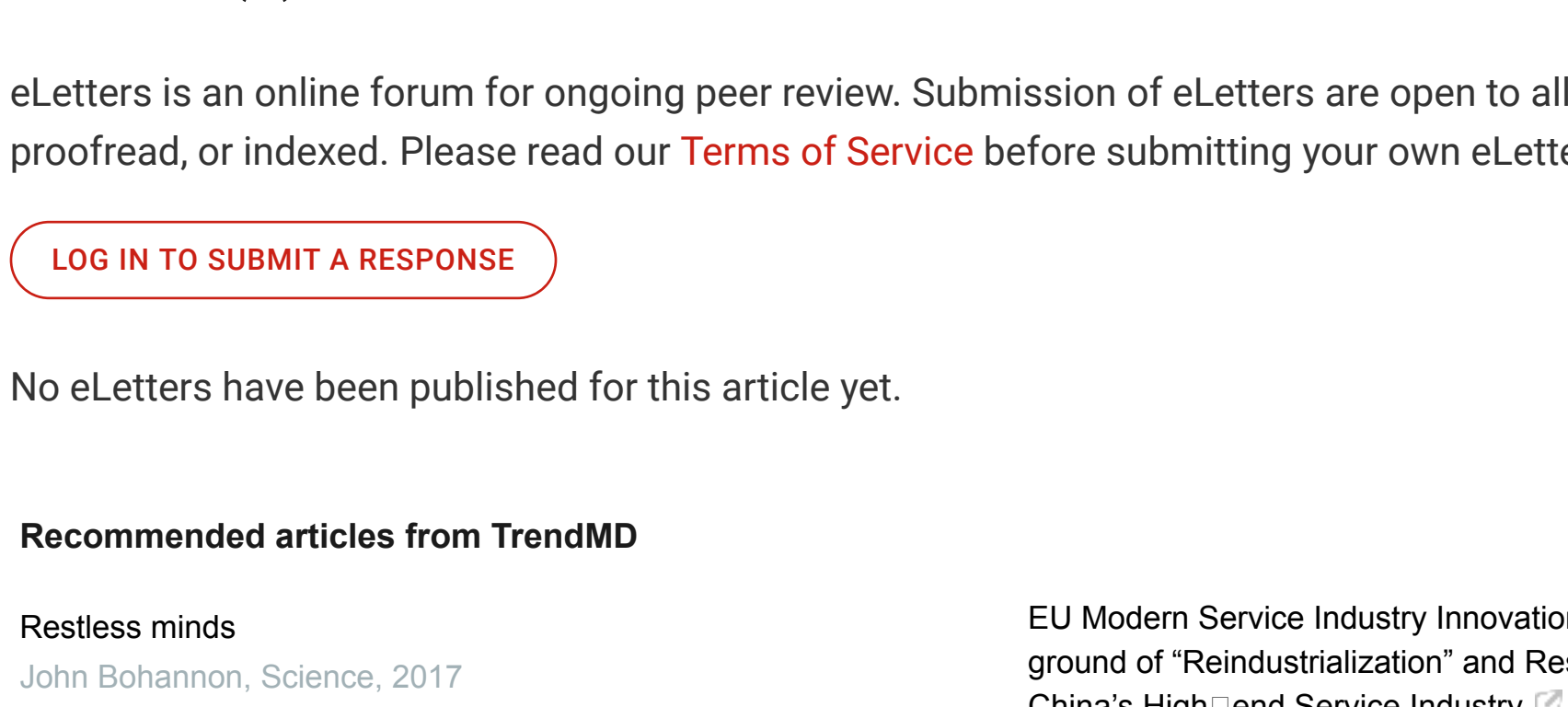
Proportion of Ph.D.s migrating from...



Residence in 2016



Annual scientific immigration to the United States



GRAPHIC: G. GRULLÓN/SCIENCE

eLetters (0)

 eLetters is an online forum for ongoing peer review. Submission of eLetters are open to all. eLetters are not edited, proofread, or indexed. Please read our [Terms of Service](#) before submitting your own eLetter.

[LOG IN TO SUBMIT A RESPONSE](#)

No eLetters have been published for this article yet.

Recommended articles from TrendMD

Restless minds

 John Bohannon, *Science*, 2017

Kai Kupferschmidt's 'excellence' fund ponders changes

 Kai Kupferschmidt, *Science*, 2017

Cover stories: Visualizing scientist migrations

 American Association for the Advancement of Science, *Science*, 2017

U.K. scientists gird for future break with EU

 Erik Stokstad, *Science*, 2017

After Brexit, U.K. scientists face a long road to mend ties with Europe

 Kai Kupferschmidt, *Science Now*, 2020

 EU Modern Service Industry Innovation against the Background of "Reindustrialization" and Research on Developing China's High-End Service Industry [✉](#)

 CHEN Jun et al., *Applied Mathematics and Mechanics* (English Edition), 2015

 Stochastic and upscaled analytical modeling of fines migration in porous media induced by low-salinity water injection [✉](#)

 Yulong Yang et al., *Applied Mathematics and Mechanics* (English Edition), 2020

 Analysis of water resources and their utilization in countries targeted by the "Belt and Road" initiative [✉](#)

 LIU Zhen-wei et al., *Arid Zone Research*, 2020

 The influences of family background and structural factors on children's academic performances: A cross-country comparative study [✉](#)

 Mengjie Lyu et al., *Applied Mathematics and Mechanics* (English Edition), 2019

 The interplay between (electro)chemical and (chemo)mechanical effects in the cycling performance of thiophosphate-based solid-state batteries [✉](#)

 Jun Hao Teo et al., *Materials Futures*, 2022

 Powered by **TRENDMD**

 I consent to the use of Google Analytics and related cookies across the TrendMD network (widget, website, blog). [Learn more](#)

CURRENT ISSUE


BTG1 mutation yields supercompetitive B cells primed for malignant transformation
 BY CORALINE MLYNARCZYK, MATT TEATER, ET AL.

CRISPR technology: A decade of genome editing is only the beginning
 BY JOY Y. WANG, JENNIFER A. DOUDNA

Heritable defects in telomere and mitotic function selectively predispose to sarcomas
 BY MANDY L. BALLINGER, SWETANSU PATTNAIK, ET AL.

[TABLE OF CONTENTS >](#)

LATEST NEWS

NEWS 24 JAN 2023
'Incredibly concerning': Bird flu outbreak at Spanish mink farm triggers pandemic fears
SCIENCEINSIDER 24 JAN 2023
Human geneticists apologize for past involvement in eugenics, scientific racism
NEWS 23 JAN 2023
When deer disappear off the menu, hungry wolves turn to sea otters
SCIENCEINSIDER 24 JAN 2023
Despite opposition, Japan may soon dump Fukushima wastewater into the Pacific
SCIENCEINSIDER 23 JAN 2023
Women scientists at famed oceanography institute have half the lab space of men
SCIENCEINSIDER 23 JAN 2023
Can California's floods help recharge depleted groundwater supplies?

ADVERTISEMENT



ADVERTISEMENT



Challenge #147

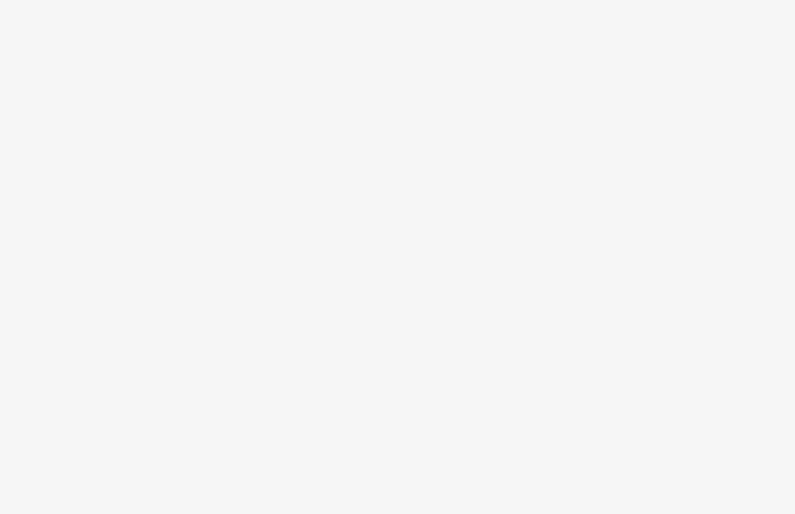
Architects of the energy transition.

We conduct research to ensure that the energy transition succeeds.

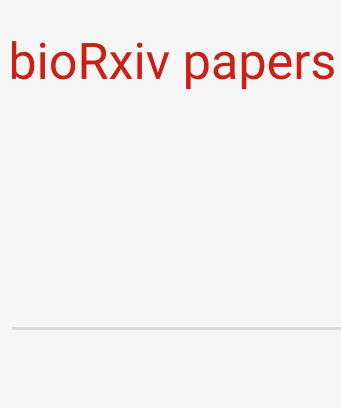
Inspired by challenges.

[Learn more](#)

ADVERTISEMENT



RECOMMENDED



30 MAR 2007 | BY ALAN KOTOK

Introducing GrantsNet 2.0

NEWS OF THE WEEK | MARCH 2011

Newsmakers

IN DEPTH | DECEMBER 2017

Blockade harms Qatari science

IN DEPTH | MARCH 2017

Raising the drawbridge

bioRxiv papers

26 APR 2017 | BY JOCELYN KAISER

BioRxiv preprint server gets funding from Chan Zuckerberg Initiative
[View full text](#) • [Download PDF](#)

 Science
Signaling

 Science
Translational Medicine

Science

 Science
Advances

 Science
Immunology

FOLLOW US



NEWS

 All News
 ScienceInsider
 News Features
 Subscribe to News from Science
 News from Science FAQ
 About News from Science

CAREERS

 Careers Articles
 Find Jobs
 Employer Profiles

COMMENTARY

 Opinion
 Analysis
 Blogs

JOURNALS

 Science
 Science Advances
 Science Immunology
 Science Robotics
 Science Signaling
 Science Translational Medicine
 Science Partner Journals

AUTHORS & REVIEWERS

 Information for Authors
 Information for Reviewers

LIBRARIANS

 Manage Your Institutional Subscription
 Library Admin Portal
 Request a Quote
 Librarian FAQs

ADVERTISERS

 Advertising Kits
 Custom Publishing Info
 Post a Job

RELATED SITES

 AAAS.org
 AAAS Communities
 EurekAlert!
 Science in the Classroom

ABOUT US

 Leadership
 Work at AAAS
 Prizes and Awards

HELP

 FAQs
 Access and Subscriptions
 Order a Single Issue
 Reprints and Permissions
 TOC Alerts and RSS Feeds
 Contact Us

AAAS